User datasheet · for firmware version 1.0.3

ByteRelay User Datasheet

Set up the controller, switch relays from the panel, buttons and API, and know how it protects your equipment.

Firmware 1.0.3 | 8 relays | Ethernet | English · فارسی

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A network-controlled relay controller for buildings

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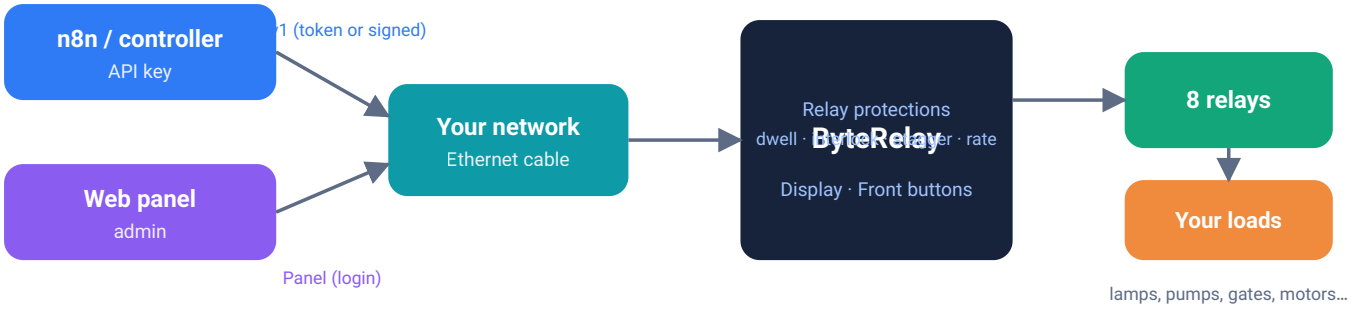
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1 What is ByteRelay

A network-controlled relay controller for buildings

ByteRelay switches up to **8 relays** from a web panel, from the two buttons on the device, from schedules and from other systems (such as n8n or your own controller) through a secure API. It is built to run 24 hours a day, all year, and to recover by itself without damage.



How the parts talk to each other

What the device does

- Switches 8 relays: on, off, toggle, or a timed pulse.
- Protects your equipment: minimum on/off time, interlock for two relays (up/down), staggered switching, a limit on switchings per minute.
- Remembers state after a power cut: each relay can start Off, On or “as before”.
- Scenes (up to 8) and weekly schedules (up to 16).
- A secure API with separate keys, each limited to chosen relays and actions.
- Shows everything on its small display and with colour lights.

What is in the kit

Part	Job
Control board	The brain of the device
Network connection (Ethernet)	Network over a cable
8-channel relay board	Switches your loads
Small display	Shows the relays and messages
Two buttons with lights	Choose a relay / switch it
Status light	Colour light for the state

i Wiring and technical details are in the **Developer datasheet**. This datasheet is for people who set up and use the device.

2 Getting started

From the box to the first switched relay in 10 minutes

1. **Connect.** Plug an Ethernet cable into the device and your router or switch. Power the device. The display shows ByteRelay and then the relays.

2. **Find the device.** The address is shown at the top of the display. You can also open `http://byterelay.local` on the same network.

3. **Sign in.** User `admin`, password `admin`. The panel then asks you to choose a new password (at least 8 characters).

4. **Name your relays.** Open **Relays** and give each relay a clear name (up to 16 characters). Choose what each relay does after a power cut.

5. **Test.** On the **Dashboard** press a relay. The big button turns green and the relay clicks.

6. **Set the time zone** in Settings. Schedules use it.

! Change the default password in step 3. If you forget it, hold both front buttons for 10 seconds to bring back admin / admin (see “Recovery and reset”).

✓ ByteRelay talks only through the network cable (LAN). It has no Wi-Fi at all: the radio is never switched on.

3 The display, buttons and lights

Everything you can do without a computer

Using the two buttons

Button	Action	Result
Left (select)	Short press	Choose the next relay (1 → 8 → 1)
Left (select)	Hold 0.7 s	Choose the previous relay
Right (switch)	Press	Switch the chosen relay on or off (when you release the button)
Both buttons	Hold 10 s	Reset the panel login to admin / admin (if allowed in Settings)

When the display is off, the first press only wakes it up, so nothing is switched by accident. The left button light breathes slowly when everything is fine and blinks fast when there is a problem. The right button light is on when the chosen relay is on.

Screens



Main screen

Eight cells: a filled cell is a relay that is on. The frame marks the chosen relay. The top line rotates between the name, the address and the time. The bottom line shows the chosen relay and its state.



Waiting

A switching is waiting for a protection delay (for example the minimum time).



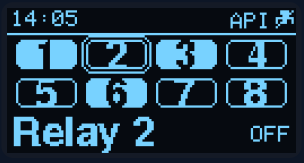
Locked

This relay is locked in the panel, so nobody can switch it on.



Notice

A short message at the bottom, for example when switching too fast.



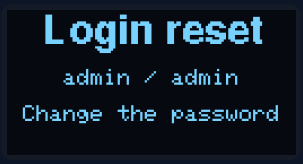
No cable

The network cable is not connected. The relays still work from the buttons and keep their schedules.



Login reset

Both buttons are held; the bar fills in 10 seconds.



Login reset done

The login is admin / admin. Choose a new password in the panel.



Updating

A firmware update is running. Do not switch off the power.

The colour light

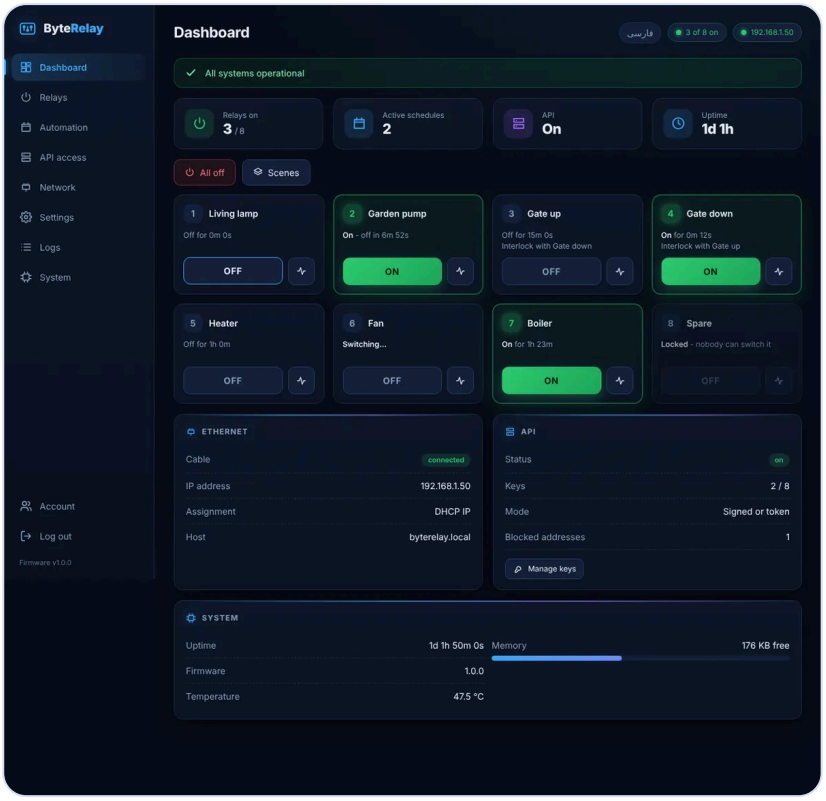
Colour	Meaning
Blue, slow breathing	Ready, network connected
Green	At least one relay is on
Amber	No network (cable or address)
Red, blinking	A problem: see the System page

4 The web panel

A tour of every page

Dashboard

One tile per relay. The big button switches it, the small button sends a pulse of the length set for that relay. “All off” switches every relay off. The page refreshes by itself every two seconds without jumping.

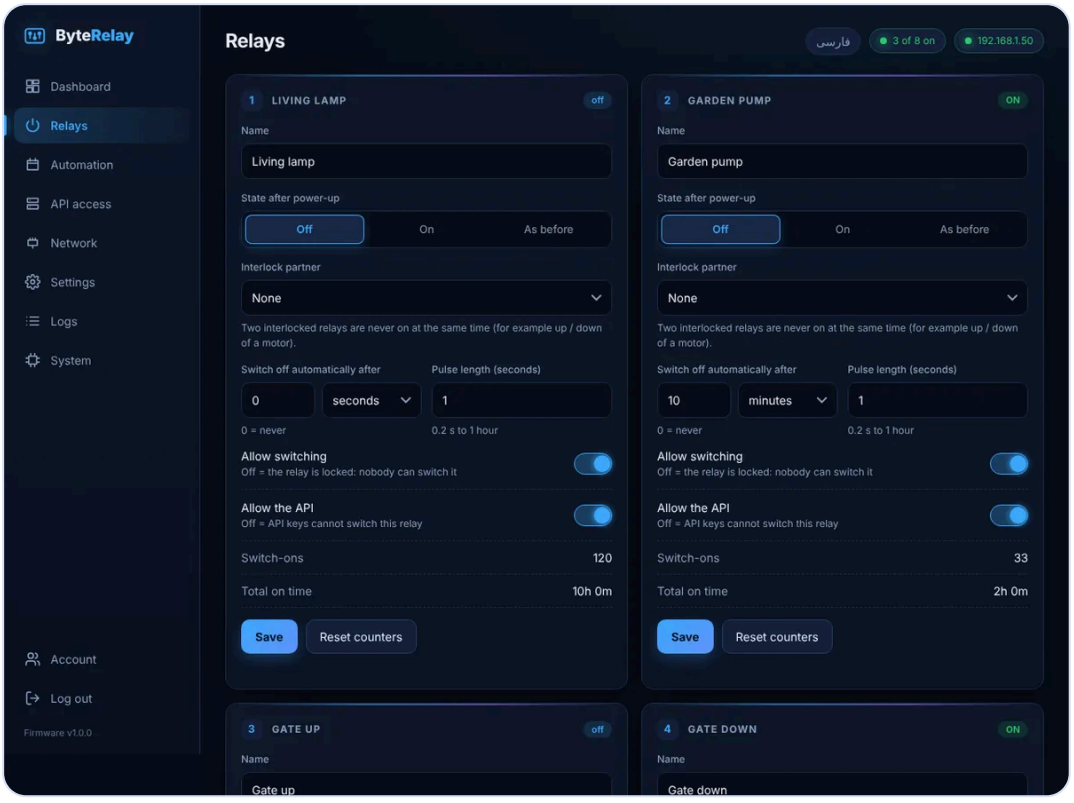


The dashboard.

Relays

Each relay has its own card.

Setting	What it does
Name	Up to 16 characters. Shown on the display, in the panel and in the API.
State after power-up	Off, On, or “as before” (the state before the power went).
Interlock partner	Two relays that must never be on together (for example a motor up / down). Choosing one partner sets it on both relays.
Switch off automatically	The relay switches itself off after this time. 0 = never.
Pulse length	From 0.2 s to 1 hour. Used by the pulse button and by default in schedules and the API.
Allow switching	Off = locked: nobody can switch it on (switching off is always possible).
Allow the API	Off = no API key can touch this relay.
Counters	Number of switch-ons and total on time: useful for maintenance.

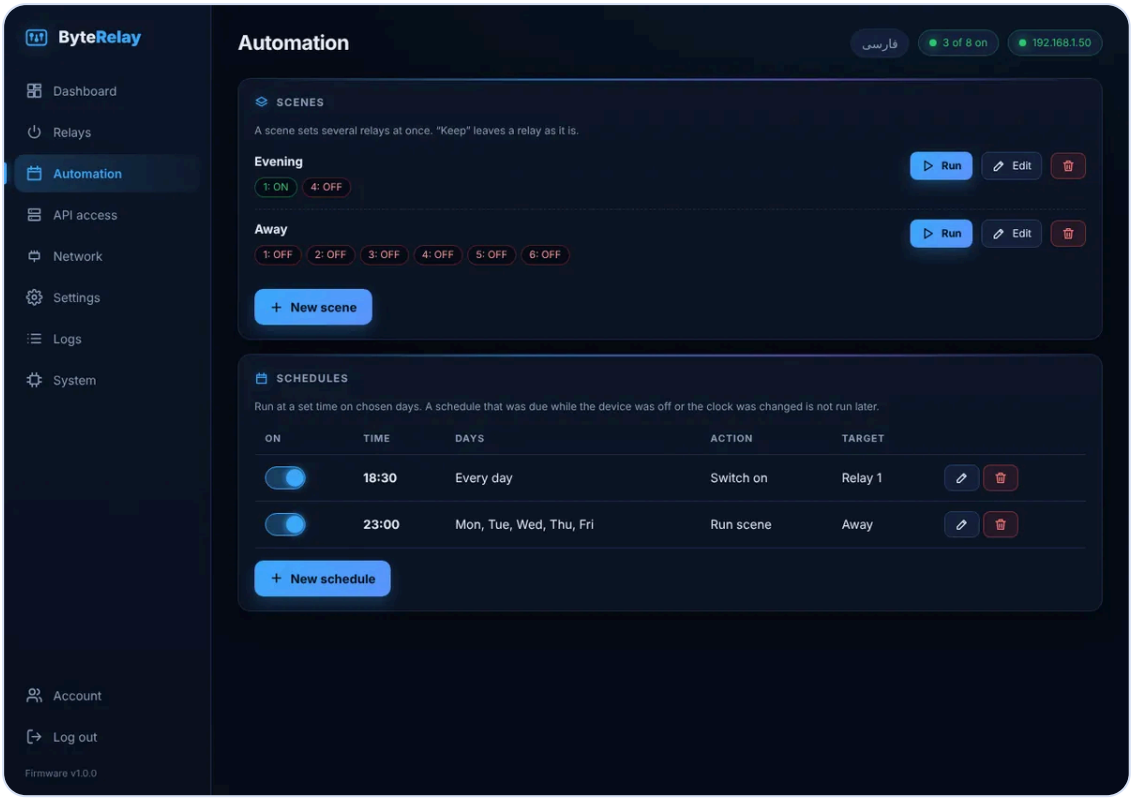


The relay cards (top of the page).

Automation

Scenes set several relays in one step: for each relay choose Keep, On or Off. A scene cannot switch on two interlocked relays together. **Schedules** run an action at a time on chosen days: switch off, switch on, toggle, pulse or run a scene.

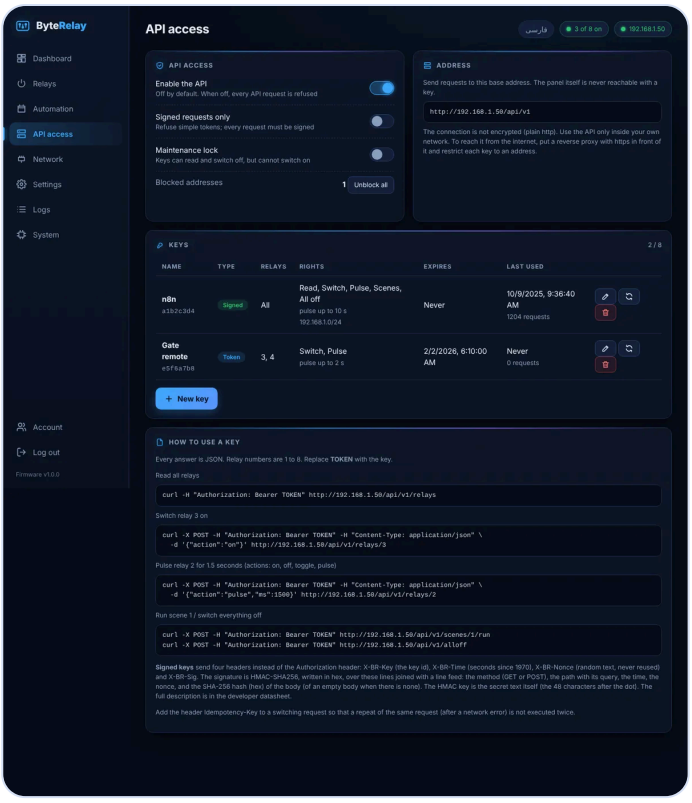
> A schedule that was due while the device was off, or while the clock was being corrected, is not run later. This is on purpose: a gate must not open by itself minutes after the power returns.



Scenes and schedules.

API access

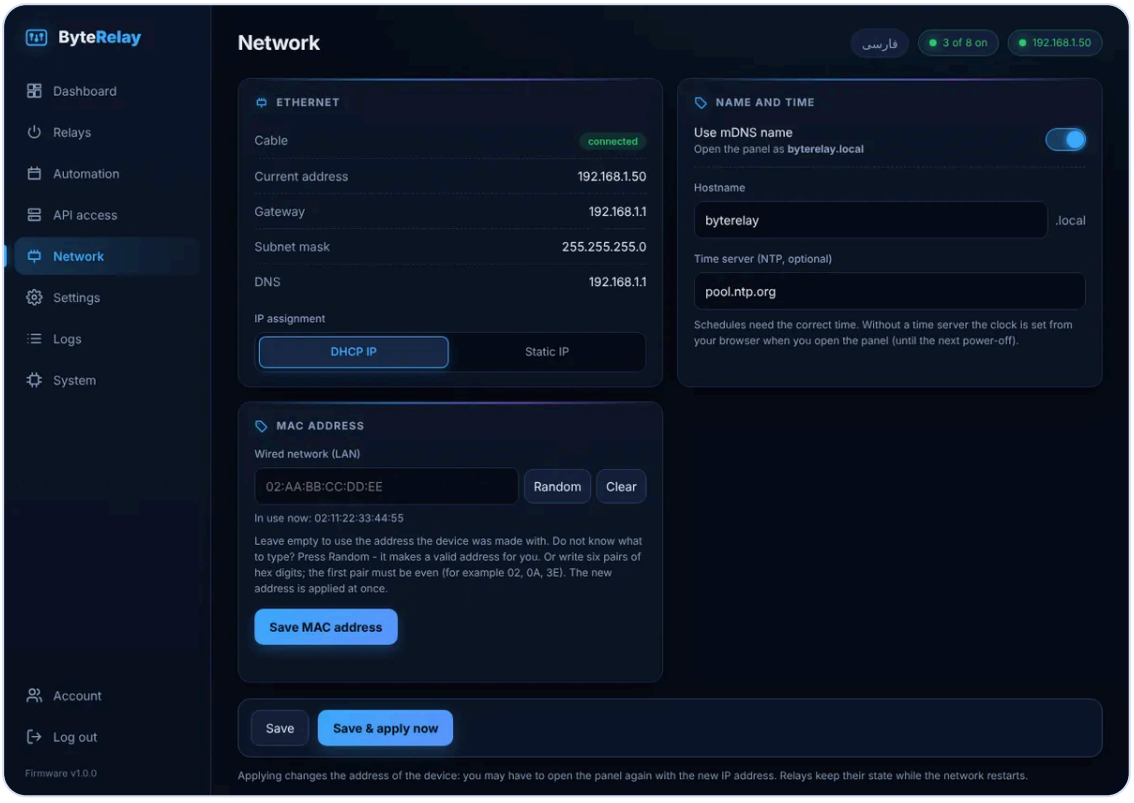
Create keys for other systems, see “Using the API”. The panel never accepts an API key, and an API key never opens the panel.



API settings and keys.

Network

DHCP or a fixed address, the name `byterelay.local`, the time server and the MAC address of the cable port. Applying a change may disconnect you for a few seconds; the relays keep their state.



Network settings.

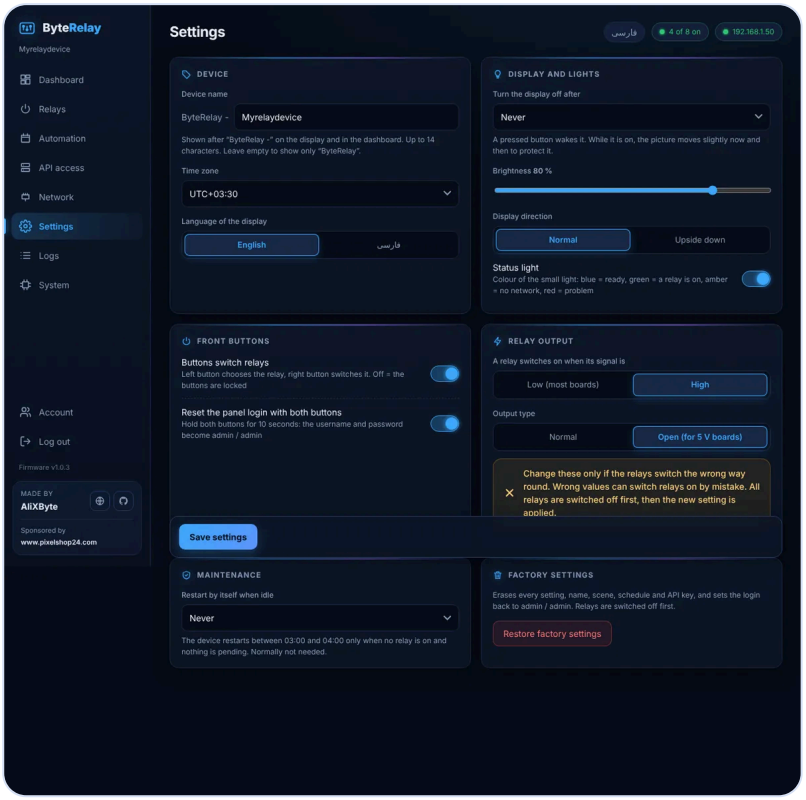
Settings

Device name, time zone, display (sleep time, brightness, direction), the buttons, the status light, the relay output type, the restart-when-idle option and factory settings.

The **device name** is a short name you choose (up to 14 characters, or empty). The device is then shown as “ByteRelay - your name” on the display and in the dashboard. The **System** page has an “About” card with the maker, the sponsor and two buttons that open the website and the GitHub page.

> Factory values: “Relay output” signal level **High** and output type **Open (for 5 V boards)**; “Turn the display off after” is **Never** (choose 1 minute … 1 hour if you want the display to sleep). If your relays work the wrong way round, change the relay output here.

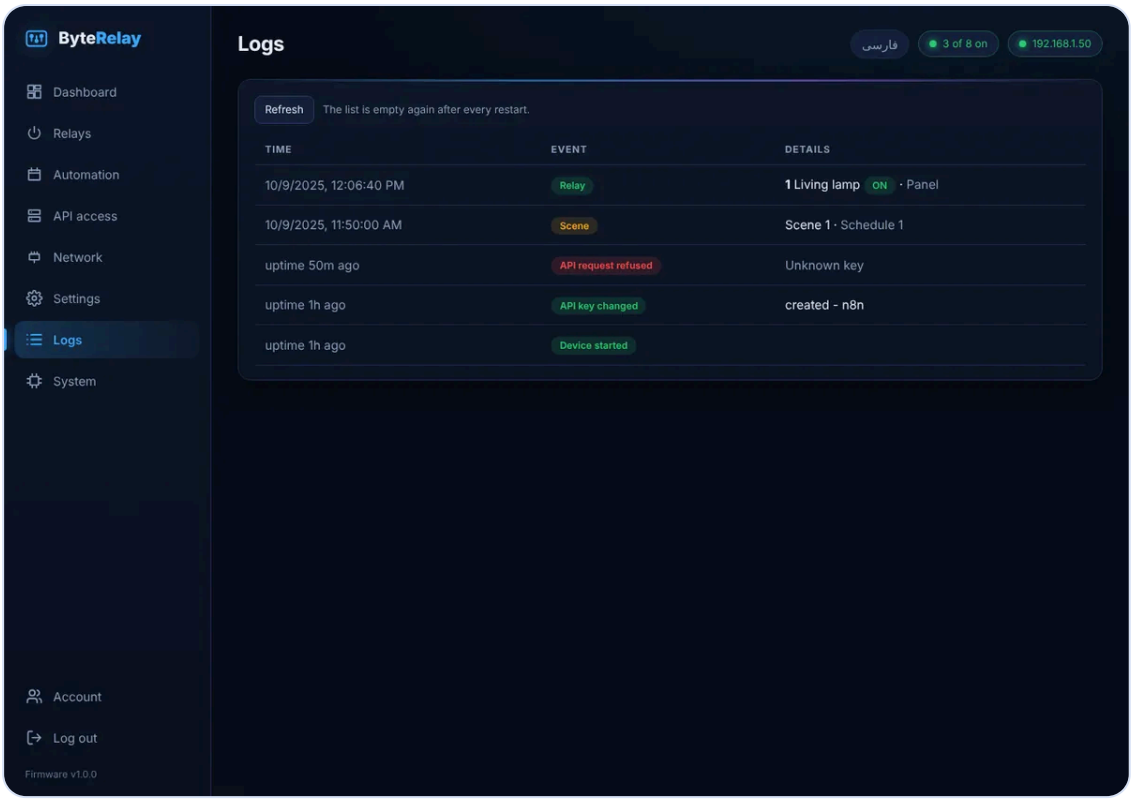
> “Relay output” (signal level and output type) must match your relay board. Change it only if the relays work the wrong way round.



Settings (top of the page).

Logs, System and Account

Logs list what happened (who switched which relay, refused API requests, logins, network changes). The list is kept in memory and is empty after a restart. **System** shows the health of every part, lets you reboot and install a firmware update. **Account** changes the user name, password and nickname.


ByteRelay
Dashboard
Relays
Automation
API access
Network
Settings
Logs
System
Account
Log out
Firmware v1.0.0
Logs
فارسی
3 of 8 on
192.168.1.50
Refresh
The list is empty again after every restart.
TIME
EVENT
DETAILS
10/9/2025, 12:06:40 PM
Relay
1 Living lamp ON · Panel
10/9/2025, 11:50:00 AM
Scene
Scene 1 · Schedule 1
uptime 50m ago
API request refused
Unknown key
uptime 1h ago
API key changed
created - n8n
uptime 1h ago
Device started

The log.

5 How the relays are protected

Why they last, and what happens when something goes wrong

Every request, from the panel, the buttons, a schedule or the API, only *asks* for a state. One dedicated task applies it and enforces the rules below, so no program can damage the relays or the equipment connected to them.

Protection	Value	Why
Minimum time in a state	0.8 s	Stops contact chatter and coil stress
Interlock pair	0.6 s pause	The first relay is released and settles before the second is energised
Staggered switching	0.2 s (0.3 s for 20 s after power-up)	Limits the inrush on the supply when many relays start
Rate limit	20 switchings per minute per relay	A faulty program cannot hammer a relay
Output read-back	every 20 ms	If the output level is ever found wrong it is corrected and logged
Automatic restart of the relay task	after 10 s without a heartbeat	If the task ever stalls it is restarted by itself
Locked relay	—	A locked relay cannot be switched on by anything; switching off is always allowed

Power cuts and restarts

- At power-up every relay output is driven to “off” as the very first action, before the network or the display start.
- Then each relay returns to its power-up setting (Off, On or as before), one after another with a pause, so the supply is not hit by all relays at once.
- A timed pulse is never restored after a restart.
- Two interlocked relays are never both restored to On.

> ! A software restart (for example after a fault is detected, or a reboot from the panel) cannot hold the outputs. The relays release for about 1–2 seconds and then return to their power-up state. If a load must never be interrupted, give that relay the setting “As before” and design the load so that a short release is harmless.

Self-recovery

- Network: if the cable is unplugged and plugged back, or the router restarts, the device reconnects by itself. Relays and schedules do not depend on the network.
- Display: if the display stops answering, the device restarts its connection by itself.
- Watchdog: a hung program makes the device restart in seconds.
- Safe mode: after several restarts in a row the optional features (schedules, API) are paused until the device runs stably.
- Firmware update: if a new version does not start properly the device returns to the previous one.

6 Using the API

Let n8n, a controller or your own program switch relays

The API is **off** until you switch it on in **API access**. Then create one key for each program. A key is limited: you choose its relays, its rights, how long a pulse it may ask for, when it expires, from which address it may be used and how many requests per minute it may send.

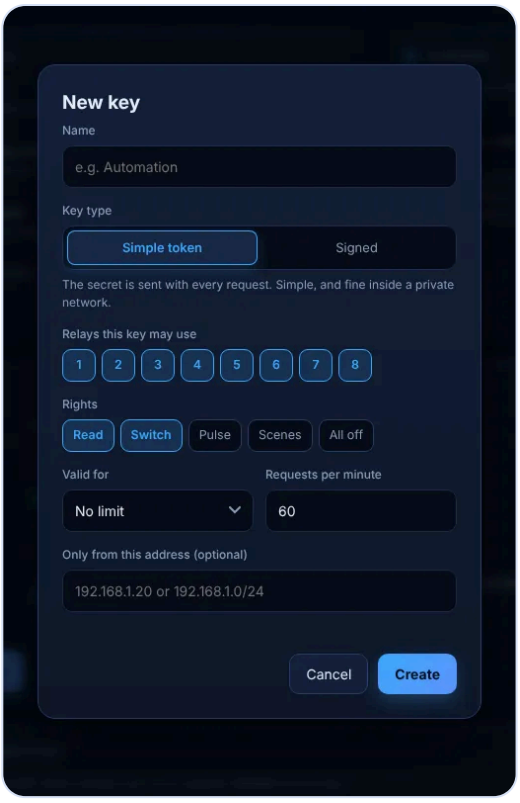
Two kinds of key

Kind	How it works	Use it for
Simple token	One secret sent in the header `Authorization: Bearer …`. Easy to use in n8n.	Private network, quick set-up
Signed	The secret never travels. Each request carries a signature that is valid for one minute and only once, so it cannot be copied or replayed.	Anything where traffic may be seen, controllers you program yourself

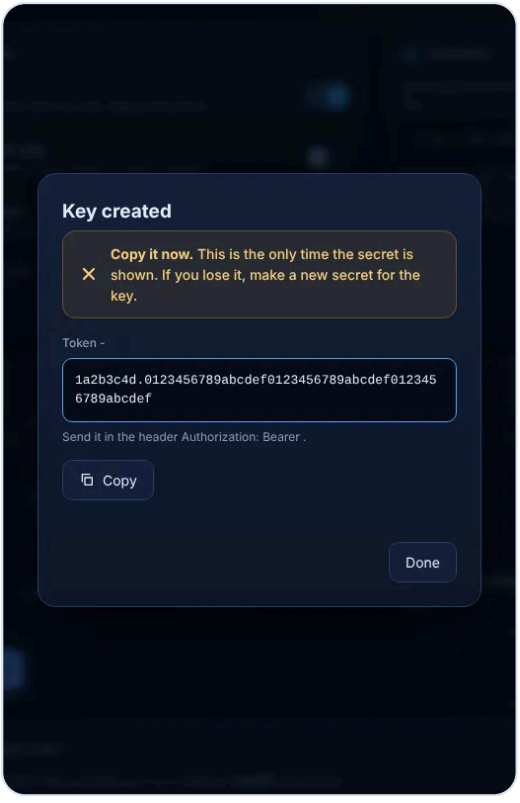
! The device speaks plain http (no https). A signed key protects against copying and replaying, but the traffic itself is not hidden. Use the API inside your own network. To reach it from the internet, put a reverse proxy with https in front of it and restrict each key to an address.

Create a key

1. Open **API access** and switch **Enable the API** on.
2. Press **New key**, choose a name, the key type, the relays and the rights.
3. Press **Create** and copy the token now. It is shown only once. Lost? Use the “new secret” button of the key.



Creating a key.



The token is shown once.

n8n example

Use the **HTTP Request** node:

Field	Value
Method	POST
URL	http://byterelay.local/api/v1/relays/3
Authentication	Header Auth: `Authorization` = `Bearer <token>`
Body (JSON)	{"action":"on"}
Header (optional)	`Idempotency-Key` = a unique value, so a retry never switches twice

The answer says `"settled": true` when the relay really reached the requested state. If a protection delay is running the answer is `202` (accepted, still in progress).

Safety of the API

- Wrong keys are counted: 5 failures from one address block it for 1 minute, then 2, 4, … up to 16 minutes. Too many failures from everywhere lock the API for a short time. “Blocked addresses” in the panel can be cleared with one button.
- **Maintenance lock** lets keys read and switch off but not switch on (for example while you work on the equipment).
- Switching everything off with the API is always allowed, even when locked.
- The relay protections of the previous chapter apply to the API as well.

7 Recovery and reset

If you forget the password or cannot reach the panel

Situation	What to do
Forgot the panel password	Hold **both front buttons for 10 seconds** (the display shows a bar). The login becomes admin / admin. Choose a new password at once. This works only if “Reset the panel login with both buttons” is on in Settings. Relays are not touched.
Forgot the network address, or a wrong network setting	Hold **both front buttons for 10 seconds**. Besides the login, the wired network goes back to DHCP, name byterelay.local and the original MAC address. Then read the new address on the display.
Changed the network and lost the address	The address is on the display, or open `byterelay.local`. If it still fails, use the button reset above.
Start again from zero	Settings → Restore factory settings (needs your password). It erases everything and sets the login to admin / admin.
The relays work the wrong way round	Settings → Relay output: change the signal level, then check every relay with nothing connected to it.

Keep the default password away from the device: after the first sign-in the panel forces a new password. Write the new one down and keep it in a safe place.

8 Security advice

Short rules that keep the system safe

- Choose a strong, unique panel password and write it down somewhere safe.
- Do not expose the device to the internet directly (no port forwarding). Use a VPN or a reverse proxy with https.
- Keep the API off if you do not need it. If you need it, use one key per program, only the relays and rights it needs, and an expiry date.
- Prefer **signed** keys, and switch on “Signed requests only” when all your programs support it.
- Restrict each key to the address of its program (for example `192.168.1.20`).
- Give a relay that must never be switched remotely the setting “Allow the API” off, or lock it.
- Safety-critical loads (gates, doors, heaters) should also have their own hardware safety devices. A network controller must never be the only protection.

9 Specifications

The numbers in one table

Item	Value
Relays	up to 8 (the number in use is set when the unit is built; unused relays show a lock)
Relay name length	1–16 characters
Scenes / schedules	8 / 16
API keys	8
Pulse length	0.2 s – 1 hour
Automatic switch-off	1 s – 24 hours
Network	Ethernet 10/100 only (no Wi-Fi), DHCP or fixed address, name byterelay.local
Panel languages	English, فارسی (the button at the top of the page; a browser that has not chosen yet follows the display language)
Display languages	English, فارسی
Default login	admin / admin (must be changed at first sign-in)
Firmware update	From the panel, with automatic return to the previous version if the new one fails

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